

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following best describes a "design paradigm" as presented in the lectures?

1 / 1 point

- ☒ A structured approach to software architecture and organization that addresses common development challenges
- ☐ A set of consistent visual cues presented to the users of an application
- ☐ A set of questions asked of either colleagues or an LLM-assistant throughout development to guide a project's design
- ☐ A list of tests that a piece of software must pass in order to be considered well-designed

✔

Correct

Design paradigms help structure how software is designed and how teams work on that software to address common challenges

2. Which of the following is NOT an advantage of Configuration Driven Development (CDD)?

1 / 1 point

- ☐ Flexibility, as software can be easily adapted to changing environments and needs
- ☐ Non-technical colleagues can change an application's behavior by adjusting configuration files but leaving source code alone
- ☐ Reduced bugs, as application behavior can be adjusted without editing source code
- ☒ More secure applications, as sensitive information can be safely stored in configuration files

✔

Correct

It's not necessarily true that applications built with a CDD approach will be any more secure. Information in configuration files is just as likely to be compromised as if it was stored in source code.

3. Which of the following best describes how the json and pickle libraries and their associated file formats compare to one another?

1 / 1 point

- ☐ JSON files can only be used in a Python environment, while Pickle files can be used by a wide variety of programming languages
- ☐ json can read and write serialized data to both strings and to files, while pickle can only read and write to files.
- ☒ The json library is used for managing data written in JSON, while pickle can handle the serialization and deserialization of a wide variety of data types in Python, in particular complex objects.
- ☐ json is used exclusively when implementing configuration driven development, while pickle can be used when implementing a wide variety of design paradigms

✔

Correct

The json library manages data written in JSON while pickle can serialize a wide variety of file formats.

4. What is the primary reason you would choose to serialize a piece of data in a program?

1 / 1 point

- ☐ To cryptographically secure its contents
- ☐ To apply the configuration settings contained in that data
- ☒ To convert it to a format that makes it easier to save to disk, move, and reconstruct later
- ☐ To enable it to be processed by an LLM

✔

Correct

This is the primary reason serialization is used.

5. Which of the following is NOT a use-case an LLM could help with when trying to use a design paradigm like Configuration Driven Development when building an application?

1 / 1 point

- ☐ Brainstorming potential design paradigms that meet your project's needs
- ☐ Suggesting libraries and formats to help apply a given design paradigm
- ☐ Exploring common commands with different libraries and packages
- ☒ Applying a variety of data serialization techniques to provided pieces of data

✔

Correct

Tasks like this one should be handled by a dedicated software library